

SKILL 21

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root@PC: ~/os_lab/combined_complex_lab
root@PC:~/os_lab/combined_complex_lab# cd ~/os_lab/combined_complex_lab
root@PC:~/os_lab/combined_complex_lab# nano modular_shell_error_handling.c
root@PC:~/os_lab/combined_complex_lab# #include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <signal.h>
#include <errno.h>
#include <termios.h>

#define MAX_LINE 1024
#define MAX_ARGS 64
#define MAX_JOBS 32

typedef enum {
    JOB_RUNNING,
    JOB_STOPPED,
    JOB_DONE
} JobState;

typedef struct {
    int job_id;
    pid_t pid;
    pid_t pgid;
    int is_foreground;
    char command[MAX_LINE];
    JobState state;
} Job;

// Global Shell State
Job job_table[MAX_JOBS];
int next_job_id = 1;

pid_t shell_pgid;
struct termios shell_tmodes;
int shell_terminal;

// Safe Async Diagnostic Logger
```

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root@PC: ~/os_lab/combined_complex_lab

pid_t shell_pgid;
struct termios shell_tmodes;
int shell_terminal;

// Safe Async Diagnostic Logger
void log_error(const char *module, const char *msg) {
    fprintf(stderr, "[ERROR - %s]: %s\n", module, msg);
}
return 0; te_pipeline(args, raw_cmd, is_background); d>");ound);");\n", jid, pid);
typedef: command not found
JOB_RUNNING,: command not found
JOB_STOPPED,: command not found
JOB_DONE: command not found
-bash: syntax error near unexpected token `}'
typedef: command not found
Command 'int' not found, but there are 29 similar ones.
pid_t: command not found
pid_t: command not found
Command 'int' not found, but there are 29 similar ones.
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'car' from deb ucommon-utils (7.0.1-0.1)
  command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
  command 'phar' from deb php-cli (2:8.2+93ubuntu1)
  command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
  command 'pchar' from deb pchar (1.5-6)
  command 'chat' from deb ppp (2.4.9-1+1.1ubuntu1)
  command 'shar' from deb sharutils (1:4.15.2-9)
See 'snap info <snapname>' for additional versions.
JobState: command not found
-bash: syntax error near unexpected token `}'
-bash: //: Is a directory
Command 'Job' not found, did you mean:
  command 'mob' from snap mob-sh (4.2.0)
  command 'wob' from deb wob (0.14.2-1)
See 'snap info <snapname>' for additional versions.
Command 'int' not found, but there are 29 similar ones.
pid_t: command not found
struct: command not found
Command 'int' not found, but there are 29 similar ones.
-bash: //: Is a directory
```

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root@PC: ~/os_lab/combined_complex_lab
command 'coinst' from deb coinst (1.9.3-5)
Try: apt install <deb name>
-bash: syntax error near unexpected token `STDERR_FILENO,'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `('
struct: command not found
-bash: syntax error near unexpected token `&'
sa_tstp.sa_handler: command not found
-bash: syntax error near unexpected token `&'
sa_tstp.sa_flags: command not found
-bash: syntax error near unexpected token `SIGTSTP,'
-bash: syntax error near unexpected token `"SignalSetup",'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `SIGINT,'
-bash: syntax error near unexpected token `SIGTTIN,'
-bash: syntax error near unexpected token `SIGTTOU,'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
-bash: //: Is a directory
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `('
job_table[i].job_id: command not found
job_table[i].pid: command not found
job_table[i].pgid: command not found
job_table[i].is_foreground: command not found
job_table[i].state: command not found
job_table[i].command[0]: command not found
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `{
job_table[i].job_id: command not found
job_table[i].pid: command not found
job_table[i].pgid: command not found
job_table[i].is_foreground: command not found
-bash: syntax error near unexpected token `job_table[i].command,'
job_table[i].state: command not found
-bash: return: job_table[i].job_id: numeric argument required
```

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root@PC: ~/os_lab/combined_complex_lab
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `"\n%-6s %-8s %-8s %-12s %s\n",'
-bash: syntax error near unexpected token `"------\n"'
-----\n"'
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token `( '
-bash: syntax error near unexpected token `?'
-bash: syntax error near unexpected token `"%d]    %-8d %-8d %-12s %s\n",'
job_table[i].job_id,: command not found
-bash: syntax error near unexpected token `)'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `"\n"'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `( '
Command 'int' not found, but there are 29 similar ones.
pid_t: command not found
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `( '
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token `status'
-bash: syntax error near unexpected token `"%d]+ Terminated with exit status %d\t%s\n",'
\n",'
-bash: syntax error near unexpected token `( '
-bash: syntax error near unexpected token `i'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `"%d]+ Killed by signal %d\t\t%s\n",'
-bash: syntax error near unexpected token `( '
-bash: syntax error near unexpected token `i'
-bash: syntax error near unexpected token `}'
job_table[i].state: command not found
job_table[i].is_foreground: command not found
-bash: syntax error near unexpected token `"%d]+ Suspended (SIGTSTP)\t\t%s\n",'
-bash: syntax error near unexpected token `)'
-bash: syntax error near unexpected token `}'
job_table[i].state: command not found
-bash: syntax error near unexpected token `"%d]+ Continued\t\t\t%s\n",'
-bash: syntax error near unexpected token `)'
-bash: syntax error near unexpected token `}'
```

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root@PC: ~/os_lab/combined_complex_lab
[1] Background process initialized (PID: 12045)
error_shell> jobs

[JID]  PID      PGID      State      Command
-----
[1]    12045    12045    Running    sleep 100

error_shell> fg 1
sleep 100
^Z
[1]+  Suspended                  sleep 100
error_shell> bg 1
[1]+  Resumed in background: sleep 100 &
[1] 1984
-bash: syntax error near unexpected token `('
error_shell: command not found
error_shell: command not found
[JID]: command not found
[1]+  Exit 127                  error_shell 100 > sleep
-----: command not found
[1]: command not found
error_shell: command not found
-bash: :s^Z: substitution failed
[1]+: command not found
error_shell: command not found
[1] 2129
root@PC:~/os_lab/combined_complex_lab# [1]+: command not found
```